

Trustworthy Explanations for Student Dropout Risk: Comparing Black-Box and Interpretable Models

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1. Introduction, Aim and Research Questions

Early identification of dropout risk can enable timely academic or financial support, but an unreliable explanation can misdirect scarce advising resources or produce unjustified conclusions about a student. This project will examine first-year undergraduate dropout risk using structured institutional data and TabICL, a peer-reviewed tabular foundation model (TFM) for in-context learning (Qu et al., 2025). TFMs achieve competitive results on small tabular datasets (Hollmann et al., 2025), but explanation quality is multidimensional and must be evaluated quantitatively and with users, not through anecdotal plots alone (Nauta et al., 2023). The principal contribution will therefore be evidence about the reliability and practical value of local explanations. Three questions will guide the study: **RQ1**: does TabICL add predictive value over simpler baselines; **RQ2**: are its local explanations faithful and stable; **RQ3**: do they help users anticipate model outputs and identify feasible support actions?

All predictive modelling and explanation evaluation will use one openly licensed UCI dataset. Short, consented written responses from the human evaluation will form the unstructured component and will remain linked to the study cases.

2. Decision Context and Intended User

The explanation will be designed for a university student-support officer reviewing currently enrolled first-year students immediately after first-semester results are finalised and before second-semester support decisions. For a student flagged as high risk, the officer will decide whether to prioritise that student for an outreach meeting and referral to appropriate academic or financial support. Missing a genuinely at-risk student may prevent timely contact, whereas excessive flagging consumes limited support capacity and can erode trust. The explanation must therefore clarify the model's output without presenting associations as causal diagnoses or recommendations.

3. Data Sources

Primary modelling data. ‘Predict Students’ Dropout and Academic Success’ (Realinho, Vieira Martins, Machado, & Baptista, 2021), UCI Machine Learning Repository dataset 697, is distributed under CC BY 4.0 and documented in a peer-reviewed data paper (Realinho et al., 2022). It contains 4,424 student records and 36 predictors spanning demographic, socio-economic, admission, administrative, academic and macroeconomic information, with labels Dropout, Enrolled and Graduate.

Prospective qualitative data. A voluntary convenience sample of eligible adult participants, primarily from the AIT community and supplemented by external volunteers where permitted and feasible, will provide brief written rationales, suggested support actions and comments on missing information while evaluating pseudonymised UCI cases. Participant source and role will be recorded so that any differences in the sample can be reported transparently. These consented, anonymised responses will form the project's unstructured dataset, will remain linked only to study case IDs, and will be analysed qualitatively for comprehension, actionability and information gaps. No student records will be collected, and responses will not enter the dropout model or be presented as explanations of real outcomes.

4. Prediction Task and Study Design

The primary prediction task will be binary: Dropout (1) versus Graduate (0). Enrolled records will be excluded as unresolved outcomes rather than confirmed non-dropout cases; this will simplify interpretation but may introduce selection bias and limit applicability to similar unresolved students. A meaningful share of Dropout records (17.3%, versus 3.4% of Graduate records) show no first-semester evaluations, so some labelled dropouts may have left before the decision point rather than after it; this timing ambiguity will be reported as an explicit eligibility limitation. The resulting class distribution will be documented. Subject to time and class support, sensitivity analysis will retain Enrolled as a separate class or group it with non-dropout, while primary conclusions will stay tied to the pre-specified binary task.

The prediction time will be immediately after first-semester results. Only variables available by then will enter the model: enrolment/admission records, demographic and socio-economic fields, administrative status, macroeconomic indicators, and first-semester outcomes. Second-semester curricular variables will be excluded as temporal leakage. First-semester grades and approved units are predictive but already historical, so they will be treated as immutable in counterfactual recourse.

5. Proposed Methodology

5.1 Predictive Modelling and Evaluation

A fixed, stratified 70/15/15 train/validation/test split will support identical model comparisons. Before any test access, protocol 1.2 froze the split, seed, target, feature windows, capacity policy and evaluation rules. TabICL will be compared with logistic regression, a depth-3 decision tree, XGBoost and a main-effects Explainable Boosting Machine (EBM). Preliminary validation-only results established feasibility: TabICL achieved ROC-AUC 0.952 and EBM 0.946, while the other baselines ranged from 0.923 to 0.945. This small validation difference does not establish TabICL's superiority and strengthens the case for comparing black-box and intrinsically interpretable models. The decision policy will rank each cohort and flag the top 20% using stable tie-breaking rather than a universal probability threshold. After all model and explanation choices are frozen, one final same-test-set evaluation will report discrimination, capacity-based recall and precision, calibration, runtime and paired 2,000-resample bootstrap intervals for model differences. The 20% capacity will remain the pre-specified primary operating point. If time permits, supplementary analysis may examine alternative plausible capacity levels without changing the primary evaluation.

5.2 Explanation Method and Faithfulness Evaluation

As a TFM is not a tree ensemble, TreeSHAP exactness will not be claimed. Model-agnostic permutation SHAP (Lundberg & Lee, 2017) will explain dropout probability using a training-only background and a feature-count-derived permutation budget. Signed member attributions will be summed within each semantic group, preserving additivity, before groups are ranked by absolute total. Deletion will mask those same groups with training references, recompute derived variables, and compare the cumulative absolute probability change with 50 random group orders; the primary statistic will be ranked-minus-random area with case-level uncertainty. Similar curves will indicate inadequate fidelity, and masking will measure reliance, not intervention. Validation checks reconstructed TabICL probabilities within $4.1e-8$ and recovered known white-box attribution ranks with Spearman 1.0. Grouped permutation importance will supply global reliance, while ALE will examine three validation-selected numerical effects with less off-manifold risk than PDP under correlation.

5.3 Explanation Stability Evaluation

The stability evaluation will deterministically select four cases from each of five predicted-risk strata (20 total) without labels. For each case, the explainer will run 10 times, varying seed and training-background sample separately. Mean pairwise top-five overlap—shared features divided by five—will be primary, with rank correlation secondary. Continuous perturbations will draw Gaussian noise with standard deviation equal to 1% of the feature's training standard deviation, clip to training bounds, recompute dependencies, and qualify only when absolute probability change is at most 0.01. Mean overlap below the validation-frozen 0.80 warning level will be displayed, not hidden through selective reruns (Tiukhova et al., 2024).

5.4 Human Evaluation

Following forward-simulation evaluation (Hase & Bansal, 2020), the human study will aim to recruit 8–10 voluntary AIT participants. At least four participants will be required, and the achieved sample will be allocated as evenly as practicable between groups using seeded assignment. After standardised instructions and one unscored practice case, both groups will receive the same label-free panel of five flagged and five unflagged profiles in random order; the explanation group will additionally see feature contributions. Neither group will receive the output, probability, capacity boundary or correctness feedback until all 10 responses are locked. The interface will record predictions, confidence, time and written responses. Participant-level accuracy will be primary; group accuracy and time will be compared descriptively alongside illustrative paraphrased comments and coded responses (Section 5.7). Recruitment shortfalls and unequal group sizes will be reported transparently. Unless support professionals participate, the study will test proxy-user comprehensibility, not professional effectiveness, and will make no significance claim.

5.5 Actionability and Counterfactual Analysis

Before modelling, features will be classified by temporal availability and feasibility. Age, nationality, prior qualifications and completed first-semester outcomes will remain fixed. Potentially resolvable tuition or debt conditions may change only when realistic at the decision point, distinguishing institutional support from student-controlled action. For a flagged case, the search will find the smallest feasible change that moves it outside the cohort's top-20% set after recomputing its rank while all other scores stay fixed; no universal threshold will be assumed, and no-recourse cases will be explicit. These outputs will describe model behaviour, not causal promises (Karimi et al., 2021).

5.6 System Architecture, Workflow and Expected Outputs

The end-to-end workflow will produce a frozen comparison of TabICL and its baselines, quantitative explanation evidence, constrained counterfactuals, subgroup diagnostics, human-evaluation findings, a completed model card and reproducible report artefacts. A lightweight browser-based research interface will provide two controlled modes: a decision-support demonstration showing the frozen prediction, local explanation, reliability warnings and feasible support options, and a blinded study view

that will suppress the answer until the participant responds. Both modes will load the same versioned preprocessing and model artefacts used in evaluation as a research demonstration, not a production AIT service.

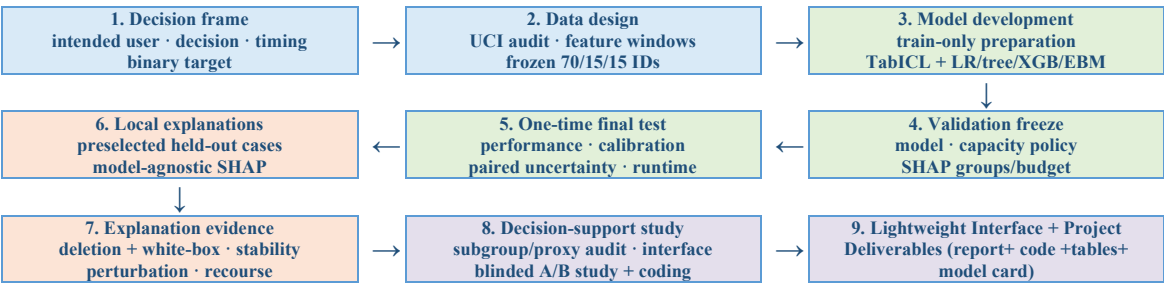


Figure 1. Proposed research workflow. Model, capacity policy and explanation protocols will be frozen on validation data before one-time final testing. Ethics, governance and reproducibility will apply throughout.

5.7 Qualitative Analysis of Human-Study Responses

After the prediction task, each participant will explain the judgement, propose a support action, and flag missing information. A pre-defined coding guide will group responses into comprehension, actionability and information-gap categories; two reviewers will independently code at least 20% of responses to report raw agreement, then resolve disagreements and report aggregate counts with paraphrased examples. This qualitative analysis will complement the forward-simulation results but will not train, validate or explain the dropout model.

6. Ethics, Fairness and Data Governance

The UCI dataset is publicly available under CC BY 4.0 with no direct identifiers; its citation and licence will be preserved, the raw file will remain unchanged, and all transformations will be documented. AIT human-study participation will be voluntary, informed-consent based, and penalty-free to decline or withdraw. No names, AIT identifiers or administrative records will be collected; raw free text will stay outside version control, accidental identifiers will be removed, and only aggregate or paraphrased findings will be reported. Before recruitment, the team will confirm with the course instructor whether institutional approval is required, obtaining it where applicable.

Protected attributes—gender, nationality, age, international status, displaced status and special needs—will be treated as explicit audit attributes. Every group count will be retained; performance metrics will be suppressed rather than silently omitted below 30 records, and unavailable comparisons will be stated. Eligible groups will receive flag rate, true- and false-positive rates, precision, mean risk and calibration gap under the exact capacity labels. All non-protected model features will be screened as potential proxies using Cramér's V for categorical features and correlation-ratio eta for continuous features. These diagnostics will be descriptive, not proof of fairness.

Plausible proxies will also be examined: parental occupation and qualification may encode socio-economic position, course and application mode may reflect demographic sorting, and debtor, tuition-fee and scholarship variables may encode financial disadvantage. Removing a protected attribute does not eliminate correlated proxy information, so influential proxies and limitations will be documented even where they weaken the model's apparent suitability. The system will support outreach prioritisation, not autonomous denial of education or services.

7. Feasibility and Model Complexity

TabICL combines learned representations with in-context information from a reference set and fails simulatability and decomposability: its reasoning cannot be recovered from a coefficient list or small tree, motivating Section 5.2's post-hoc protocol. By contrast, the main-effects EBM is intrinsically interpretable: each prediction is exactly the logistic transform of an intercept plus one learned shape-function contribution per feature. Its validation probabilities reconstruct from those terms within 3.4e-16, and global term strengths plus fixed low-, median- and high-risk local decompositions are exported. Logistic regression and the depth-3 tree provide further simulatable comparisons. TabICL's validation ROC-AUC exceeds EBM by only 0.006, so the project does not presume that complexity is operationally justified; final same-test-set performance, calibration, runtime and explanation evidence will determine whether TabICL remains a research comparison or EBM is the more defensible deployment candidate.

8. Reproducibility and Documentation

The repository separates source data, EDA, modelling, explanation evaluation, study materials and outputs. A pinned environment supports a fresh clone. Seed 42 and source-row IDs define the split; protocol 1.2, 26 in-scope tests, four baselines, capacity-consistent recourse, exact EBM decomposition, TabICL/SHAP integration, grouped importance, ALE, explicit subgroup suppression and proxy tables are implemented. Before final submission, one command will regenerate every report artefact and another will launch the interface from frozen artefacts. The completed model card will record intended use, data, final metrics, explanation evidence, subgroup results and limitations.

9. Expected Contribution, Limitations and Contingencies

The proposed contribution will extend beyond accuracy by asking whether TabICL's post-hoc local explanations merit trust and whether an intrinsically interpretable EBM offers a more defensible alternative. Same-test-set baselines, deletion faithfulness, white-box recovery, repeated-run and perturbation stability, human forward simulation, qualitative evidence and feasibility-constrained counterfactuals will form a coherent evaluation. Expected limitations include the single-institution UCI context, exclusion of unresolved Enrolled outcomes, approximate model-agnostic attribution, non-causal observational data, a small convenience sample, and likely non-professional audience proxies; therefore, the AIT study will test comprehensibility, not predictive validity or deployment suitability at AIT. If recruitment reaches only the minimum sample, the study will remain descriptive. Findings will remain valuable if EBM matches or exceeds TabICL, explanations are unstable or unfaithful, participants gain no benefit, or feasible recourse is unavailable.

Code and data availability. <https://github.com/TheInterpreters/Student-Dropout-Risk>

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